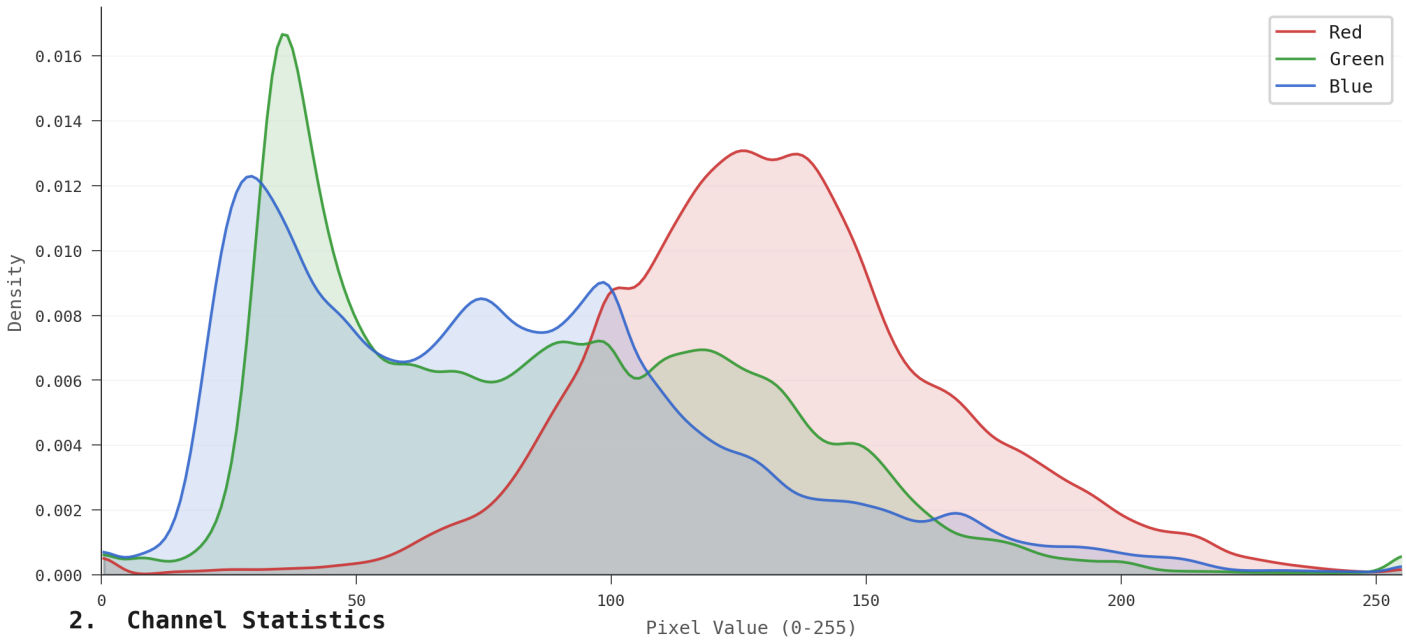
Reference Data Sheet: KODIM04

Subject: Portrait, Woman in Red Hat

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	131.8947	34.3880	1182.5358	0.6795	0.1836	0	255
Green	84.1262	43.7589	1914.8435	-0.1743	0.5638	0	255
Blue	78.3804	45.3034	2052.4016	0.3873	0.8186	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.6004	0.6850
G	0.6004	1.0000	0.9568
B	0.6850	0.9568	1.0000

Pairwise Pearson Correlation Coefficients

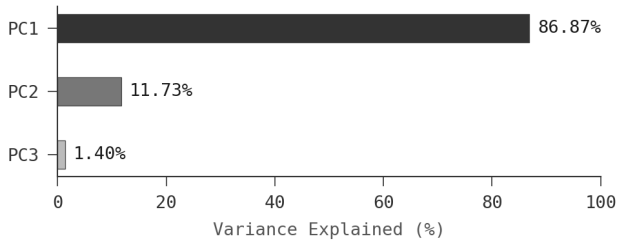
r(R-G)	=	**0.600446**
r(R-B)	=	**0.684965**
r(G-B)	=	**0.956832**
Avg |r|	=	**0.747414**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1182.5388} & 903.5425 & 1067.1069 \\
903.5425 & \mathbf{1914.8484} & 1896.8578 \\
1067.1069 & 1896.8578 & \mathbf{2052.4069}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM04

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.3905	-0.6332	**-0.6682**	4473.6423
PC2	-0.9085	0.3824	0.1685	604.2539
PC3	-0.1488	-0.6729	0.7246	71.8979

6. Derived Metrics

Condition Number

62.22

(l1/l3) PC1/PC2: 7.40 | PC2/PC3: 8.40

Blue Channel Independence

2.7%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Blue

Loading: -0.6682 | Green-Blue coupled

Classification: Weakly One-Dimensional (PC1 = 86.87%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.953348	0.957096	0.955222
Green	0.971137	0.974165	0.972651
Blue	0.972820	0.975482	0.974151

8. Average Local Variance (3x3)

Channel	Value
Red	85.9222
Green	86.4162
Blue	87.7434

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 62.22 | PC1: 86.87%

Correlation: G–B pair dominates at r = 0.956832 | R–B: 0.684965 | R–G: 0.600446

Blue Channel: 2.7% independent | 97.3% captured by PC1

Spatial: Avg autocorrelation: 0.967341 | Avg local variance: 86.6939

Eigenvector: Green-Blue coupled | G (-0.6332), B (-0.6682) co-load | R offset (-0.3905)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."